

Product Requirements Document: Smog Forecast Agent

Track: BRICS Hackathon — Track 2: Clean Air & Climate Resilience **Theme:** Sustainability **Version:** 1.0 **Status:** Draft — Hackathon Build

1. Problem Statement

Major BRICS cities monitor macro-level air quality but consistently miss early, actionable warning of pollution spikes. Existing systems are largely reactive — they report current AQI but do not forecast it, and they lack a mechanism to route that information to the specific authority who needs to act. Trans-boundary and hyper-local events (industrial emissions, agricultural burning, temperature-inversion smog buildup) are especially poorly served, because they require combining multiple data sources (ground sensors, meteorology, satellite imagery) that today live in separate systems.

The result: interventions happen after a smog event has already peaked, when the public health impact is already occurring, rather than in the 24–72 hour window when preventive action (traffic restrictions, industrial curbs, public advisories) would be most effective.

2. Goal & Vision

Build an AI-powered early-warning system that **forecasts** air quality spikes 24–72 hours in advance for a given economic corridor, automatically **drafts and routes** a localized alert to the relevant authority, and is architected so that **other BRICS nations can retrain the same pipeline on their own data** without rebuilding it — directly answering the "coordinate resources across BRICS" requirement of the track.

Vision statement: *A shared, swappable early-warning layer for BRICS cities — the same model architecture, retrainable on any nation's local data, alerting the right authority before the smog arrives, not after.*

3. Target Users

User	Need
District pollution control officer	Needs advance notice of a forecasted AQI breach in their jurisdiction, with enough context (likely cause, severity) to decide on intervention (traffic curbs, industrial notices, public advisory).
City/state environmental agency	Needs an aggregated view across all districts/corridors to prioritize resources.
Public / citizens (secondary, future scope)	Would eventually receive public advisories, but MVP focuses on authority-facing alerts first.

4. Scope

4.1 In scope for hackathon MVP

Single geography: **Delhi-NCR**, winter smog season, using publicly available historical data.

Forecasting model predicting AQI 24-72 hours ahead per district/corridor.

Rule-based threshold check against India's AQI severity categories.

AI agent that drafts a natural-language alert (English + Hindi) with likely cause and recommended action, based on forecast + feature importance.

Map-based dashboard visualizing forecasted severity by district.

A backtest demonstration against one real historical smog event.

A swappable configuration file demonstrating the pipeline could be repointed at another country's data sources and thresholds.

4.2 Explicitly out of scope for hackathon MVP

Live telephony/SMS integration to real government systems (simulated/mock routing only).

Multi-country live data integration (demonstrated via config file only, not a second live dataset).

Public citizen-facing app or notification channel.

Model retraining pipeline automation (manual retrain process is acceptable for MVP).

Handling pollutant types beyond PM2.5/PM10-driven AQI (e.g., no separate ozone or NO2 modeling).

5. Core Features

5.1 Forecasting Engine

Input: Historical AQI (CPCB), meteorological data (IMD — wind, temperature, humidity), satellite-derived features (Earth Engine — aerosol optical depth, active fire detections).

Output: Predicted AQI value (and category) per district/corridor for the next 24–72 hours, with a confidence indicator.

Implementation: Vertex AI AutoML Forecasting, with a simple trend-based fallback model as contingency.

5.2 Threshold & Alert Logic

Rule-based mapping from predicted AQI to India's official severity categories (Moderate/Poor/Very Poor/Severe).

Configurable per country via the interoperability config (see 5.4).

Triggers the agent draft step only when a breach is predicted.

5.3 AI Alert Drafting Agent

Model: Gemini, using function-calling to read structured forecast + feature-importance data.

Output: A concise, localized alert containing: affected district/corridor, predicted severity, likely contributing cause (inferred from feature importance — e.g., wind stagnation vs. fire activity), and a recommended intervention.

Localization: Draft rendered in English and Hindi via Translation API.

Routing (simulated for MVP): Alert delivered to a mock authority dashboard/webhook; architecture designed to extend to Dialogflow/SMS in future.

5.4 Interoperability Configuration

A schema-driven config (`country_config.json`) defining: feature field names, AQI threshold bands, data source labels, language for alert output.

Demonstrates that the same trained pipeline architecture could be repointed at a different BRICS nation's data without code changes — the core "shared predictive models" requirement of the track.

5.5 Dashboard

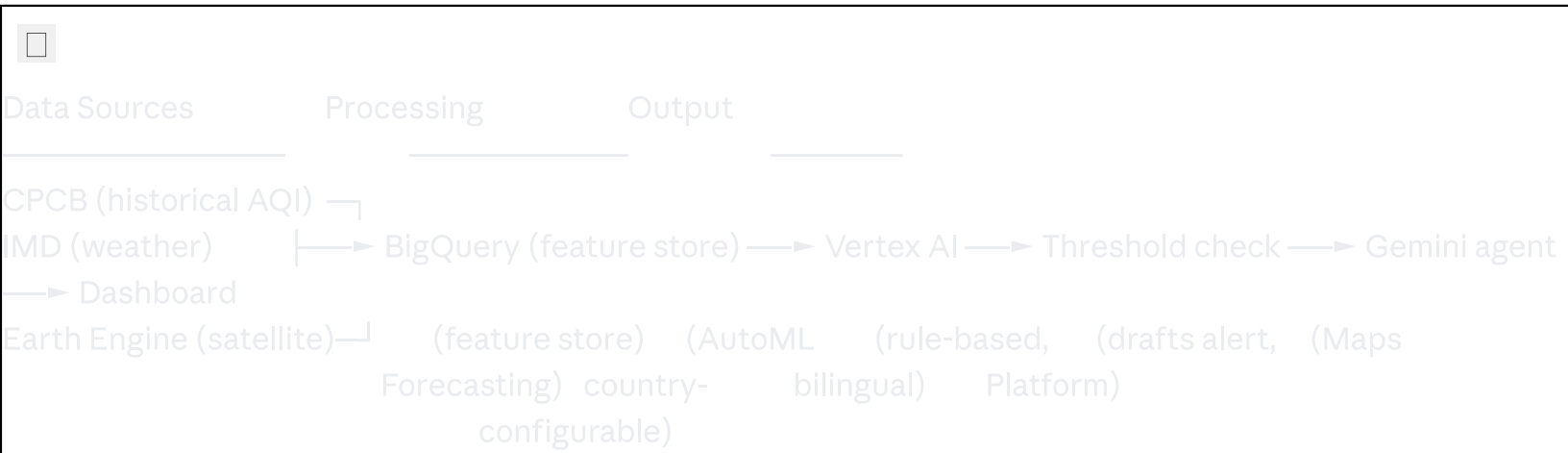
Map view (Google Maps Platform) of Delhi-NCR districts, color-coded by forecasted severity.

Click-through to view the drafted alert for a flagged district.

5.6 Backtest Validation

Pipeline run against one real historical smog spike date, forecasting "as if" 48 hours prior, to demonstrate the model would have correctly flagged the event in advance.

6. Technical Architecture



Key components:

BigQuery — central feature store; schema: `(district, timestamp, aqi, wind_u, wind_v, temp, humidity, fire_count, aerosol_depth)`

Vertex AI (AutoML Forecasting) — primary prediction model; trend-based script as fallback

Gemini (function-calling) — alert drafting agent

Translation API — bilingual alert rendering

Google Maps Platform — dashboard visualization

Earth Engine — satellite feature extraction (aerosol depth, fire detections)

6.1 Tech Stack

Layer	Tool	Purpose in this project
Generative AI & agents	Gemini API	Powers the alert-drafting agent — reads forecast + feature-importance data and generates the localized alert with cause and recommended action
Predictive modelling	Vertex AI (AutoML Forecasting)	Core AQI forecasting model, 24–72hr horizon; trend-based script as fallback
Vision & multimodal	Vertex AI Vision / Gemini multimodal	<i>Reserved for future scope</i> — citizen-submitted smoke/sky photo classification (not in hackathon MVP, noted for roadmap extension)
Language & voice	Translation API	Renders drafted alerts in English and Hindi

Layer	Tool	Purpose in this project
Geospatial	Google Earth Engine	Satellite feature extraction — aerosol optical depth, active fire detections (agricultural burning signal)
Geospatial	Google Maps Platform	Dashboard visualization — district-level choropleth showing forecasted severity
Data & backend	BigQuery	Central feature store for all ingested data (AQI, weather, satellite features)
Data & backend	Firebase	<i>Reserved for future scope</i> — auth and real-time sync if extending to a live authority-facing app
Data & backend	Cloud Run / Cloud Functions	Hosting the pipeline orchestration (threshold check → agent trigger → dashboard update)
Public data	data.gov.in (CPCB)	Historical AQI data source
Public data	IMD	Meteorological data (wind, temperature, humidity) source
Public data	ISRO/Bhuvan, Earth Engine catalog	Supplementary satellite data source alongside Earth Engine processing

Rows marked "reserved for future scope" are included to show the full platform capability but are not part of the 20-day hackathon build — see Section 4.2 for confirmed out-of-scope items.

7. Success Metrics (for demo/judging)

Metric	Target for demo
Forecast lead time	24–72 hours before threshold breach
Backtest accuracy	Model correctly flags at least 1 real historical smog event in advance
Alert quality	Drafted alert includes severity, likely cause, and recommended action — not generic text
Interoperability	Config file demonstrably swaps thresholds/schema without code changes
Language coverage	Alert renders correctly in English and Hindi

Note: these are demo-credibility metrics for a hackathon submission, not production SLAs.

8. Risks & Mitigations

Risk	Mitigation
AutoML training takes too long / underperforms	Trend-based fallback model built in parallel; start AutoML training early (Day 8 of build roadmap)
Historical yield/AQI data has gaps	Restrict demo to date ranges with confirmed clean data; inspect data early (Days 2–3)
Live telephony/SMS integration too complex for timeframe	Simulate alert routing via mock dashboard/webhook instead
Judges question single-country demo vs. "BRICS-wide" claim	Lead with the swappable config file as concrete evidence of the multi-nation design, rather than claiming live multi-country data

9. Open Questions

Should the fallback model be shown at all in the final demo, or purely a contingency? *(Recommendation: keep it hidden unless AutoML underperforms — don't dilute the pitch with a "backup" model.)*

Do we want to attempt a second (even lightly mocked) country dataset to strengthen the interoperability story, time permitting?

What specific historical smog date will we use for the backtest? *(To be locked by Day 19 of the build roadmap — should be chosen as early as data availability allows.)*

10. Appendix: Related Documents

20-Day Build Roadmap ([smog_forecast_agent_roadmap.md](#)) — day-by-day execution plan mapped to the phases in Section 5.